Luke Eyles

Mechanical and Mechatronics Engineering Student
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Education

Bachelor of Mechanical and Mechatronic Engineering, 2017 – 2022 (Expected)
Diploma in Professional Engineering Practice
University of Technology Sydney

- GPA: 6.55/7.00, High Distinction average.
- Mechanical and Mechatronic Design: Worked in a team in collaboration with Thales Australia on the Blutonomy project, researching underwater robot communication.
- Robotics: Worked in a team to program a robot arm to play tic-tac-toe with a person [video link].
- Programming for Mechatronic Systems: Programmed a simulated aircraft in C++ to intercept enemy aircraft by predicting their trajectory.
- Mechatronics 2: Built an Arduino security system using a distance sensor mounted on a motor, capable of mapping a room and detecting intruders.

Experience

Technology Consulting Intern Nov 2020 – Present
Isle Utilities

- Performing comprehensive technology scans on areas including water conservation, leakage management, and waste to energy.
- Creating communication materials for clients including reports, emails, videos, and presentations.
- Drafted trial proposals for collaborative trials of technologies.

Engineering Intern in Research and Development Department Aug 2019 – Feb 2020
Jenkins Engineering Defence Systems

- Created Signal Generator GUI application in MATLAB that interfaced with software defined radios to transmit pre-set or user defined signals.
- Developed Signal Analyser GUI in MATLAB, allowing a quick overview of signal characteristics, or in-depth analysis.
- Created system to record sensor data from an IMU to a single board computer using C++.

Assistant Lab Tutor for Introduction to Electrical Engineering Mar 2018 – Jun 2018
University of Technology Sydney

Skills

Technical Skills

- **C++:** Object oriented programming, using threads to process and send multiple streams of data, compiling using CMake.
- **MATLAB**: Image processing, signal processing, robotics using Peter Corke's Robotics Toolbox and ROS Toolbox, GUI programming using App Designer.
- **Robot Operating System (ROS)**: Controlling real and simulated robots, processing incoming messages from sensors using both Matlab and C++, visualising using RViz.
- **Microcontrollers**: Interfacing with sensors, controlling LCD screen and motors, using timers.
- **Git**: Version control and collaboration.
- **CAD***:*
    - **Solidworks**: Designing parts, performing FEA, creating assemblies, creating and interpreting technical drawings.
    - **Blender**: Creating meshes, using modifiers, applying procedural and image-based textures.
- **Python**
- **Microsoft Excel:** Recording data, using formulas, creating charts, sorting and filtering data.
- **Microsoft Office**

Core Skills

- **Teamwork and team leadership:**
    - Dividing work and delegating between team members, setting deadlines, contributing to team meetings, and ensuring all work is cohesive and completed to a high standard.
- **Visual communication skills**:
    - Drawing up diagrams to communicate concepts to others clearly in presentations, reports, and tutoring.
- **Written and verbal communication skills:**
    - Writing clear and concise reports, tailored to the audience.
    - Listening to requirements and writing formal project specification documents.
    - Participating in team meetings by contributing new ideas and adding to others' ideas.
    - Giving presentations.

References

Available on request.